

PHASE 2 DEVELOPMENT PROPOSAL

CNC Machine Monitoring System

UI/UX design · Web application · Role-based access · Program transfer · Desktop application · Mobile application (Android + iOS) · Deployment

TOTAL PROJECT COST ₹ 15,00,000 Fixed price for the defined scope	DURATION 5 months Including design & store approvals	SCOPE COVERED 44 screens 31 web + 13 mobile
---	---	--

PART A — UI/UX DESIGN & FRONTEND DEVELOPMENT

#	DELIVERABLE	COST (INR)
1	UX Research & Information Architecture User flows, sitemap, wireframes, navigation structure and overall UX strategy — the blueprint that ensures all 44 screens work as one coherent system.	₹ 50,000
2	Design System & UI Components Color palette, typography, icons, buttons, form controls, tables, charts, reusable Figma components and design tokens — guarantees visual consistency across web, desktop and mobile.	₹ 65,000
3	Web Application UI Design — 31 Screens Complete screen designs for the Dashboard, Analytics, Company Management, Program Transfer, Notifications, Profile, Settings, Maintenance and Administration modules.	₹ 1,40,000
4	Mobile UI Design & Integration — 13 Screens Android & iOS screen designs including Login, Dashboard, Machine Details, Notifications, Alarms, Profile and Settings, adapted to both platform conventions.	₹ 75,000
5	Responsive UI Development (Tailwind CSS) Pixel-perfect implementation of the approved UI using reusable Tailwind components, working across desktop, laptop and tablet resolutions.	₹ 70,000
6	Angular UI Integration Integration of all designed screens into the Angular application as reusable components with routing and state structure (UI layer only).	₹ 55,000
7	Responsive Testing & UI QA Cross-browser testing, responsiveness verification, UI fixes and final design support until sign-off.	₹ 45,000
	Subtotal — Part A	₹ 5,00,000

PART B — APPLICATION & BACKEND DEVELOPMENT

#	DELIVERABLE	COST (INR)
1	Analytics Dashboards — Backend & Real-Time Integration (9 Screens) Machine-data (MQTT) ingestion pipeline, dedicated computation services for OEE, energy, downtime and operator performance, aggregation and history APIs, and live updates wired into every dashboard so values refresh in real time.	₹ 1,80,000
2	Company & User Administration — Backend (10 Screens) Multi-company data model, CRUD APIs for companies, plans, users and roles, plan assignment and access-management business logic.	₹ 1,00,000
3	Role-Based Access Control Secure authentication, four-tier hierarchy (Super Admin → Admin → Manager / Supervisor / Operator), permission middleware enforced on every API, and role-based menus and route guards in the frontend.	₹ 65,000
4	Alarm, Ticket & Maintenance — Backend (6 Screens) Alarm capture from machine signals, ticket lifecycle APIs, downtime reason management, and preventive / periodic maintenance scheduling services.	₹ 75,000
5	Notification Engine Event-driven notification service, Firebase (FCM) push delivery, in-app notification APIs and per-user preference filtering.	₹ 55,000

6	Program Transfer Module Secure file storage, machine communication service, transfer queue, and a complete audit trail of every transfer — user, program, machine and timestamp.	₹ 85,000
7	Mobile Application Development — Android & iOS (13 Screens) Functional development of the designed screens on both platforms, API integration, push notification handling, and device testing — using the same login credentials as the web application.	₹ 2,20,000
8	Windows Desktop Application (EXE) The complete application packaged as an installable Windows program with automatic updates.	₹ 45,000
9	Play Store & App Store Publishing Store listings, screenshots, privacy documentation and complete management of the Google and Apple review processes through to public release.	₹ 45,000
10	Server Deployment & Go-Live Cloud environment provisioning — application server, database, MQTT services, SSL security and domain setup — with monitoring for reliable 24×7 operation.	₹ 55,000
11	System Testing & Quality Assurance Integration testing across web, desktop and mobile — every screen, user role and machine connection verified to ensure a stable production release.	₹ 75,000
Subtotal — Part B		₹ 10,00,000

PROJECT TOTAL

Part A — UI/UX Design & Frontend Development	₹ 5,00,000
Part B — Application & Backend Development	₹ 10,00,000
Grand Total — Phase 2	₹ 15,00,000

Please note: The above pricing covers design, development, publishing and deployment services only, and does not include third-party payments, which are payable directly by the client: Google Play Developer registration (one-time, tentatively ₹2,375 / US\$25); Apple Developer Program membership (annual, tentatively ₹9,405 / US\$99); and cloud server & hosting charges as per the selected provider's tariff. Push notifications are delivered via Google Firebase (FCM); any Firebase charges beyond Google's free usage tier, as per Google's pricing, are also payable directly by the client.

MODULE-WISE ENGINEERING DETAIL — FRONTEND & BACKEND

What is actually built for each module — every screen carries work on both sides of the system.

MODULE	FRONTEND WORK	BACKEND WORK
Analytics Dashboards 9 screens	Chart and KPI components, date/shift/machine filters, live-updating widgets, drill-down views, responsive layouts per approved design.	MQTT data ingestion, per-metric computation services (OEE, energy, downtime, operator performance), aggregation & history APIs, live push of fresh values to the screen.
Company & User Administration 10 screens	Create/edit forms with validation, list and detail views, plan assignment and access management interfaces.	Multi-company data model, CRUD APIs for companies, plans, users and roles, business rules for plan limits and access scopes.
Role-Based Access Control	Role-based menus and navigation, route guards, permission-aware buttons and screen elements per role.	Secure login & token management, four-tier role hierarchy, permission middleware checked on every single API request.
Alarm, Ticket & Maintenance 6 screens	Alarm and ticket creation forms, detail views with status workflow, downtime reason and maintenance setup screens.	Alarm capture from machine signals, ticket lifecycle APIs, preventive / periodic maintenance schedulers running automatically.
Notification Center 3 screens	Notification list with unread indicators, detail view, preference controls.	Event-driven notification engine, Firebase (FCM) push delivery, read/unread tracking, per-user preference filtering.
Settings & Profile 2 screens	Profile and application settings forms with validation.	Profile APIs, secure password change, settings persistence per user.
Program Transfer 1 screen	Program upload interface, machine selector, transfer status and history view.	Secure file storage, machine communication service, transfer queue, full audit trail (user, program, machine, timestamp).
Mobile Application 13 screens, Android & iOS	All designed screens built for both platforms — login flows, live dashboards, machine details, alarms, notifications, profile & settings.	Mobile-optimised APIs, token-based authentication shared with web, push notification registration and delivery per device.

DETAILED SCREEN INVENTORY

Web Application — 31 Screens

MODULE	SCREENS
Analytics Dashboards	Overall Dashboard, Maintenance Dashboard, Energy Dashboard, Alarm Report Dashboard, Operator Performance, OEE Dashboard, Preventive Maintenance, Periodic Maintenance, Downtime Reason Analysis
Program Transfer	Program Transfer
Company & User Administration	Company Management, Create Company, Company Dashboard, Assign Plan, View Plan, Manage Access, User Management, Create User, Roles & Permission, Create Role
Alarm, Ticket & Maintenance	Create Alarm, Alarm Details, Create Ticket, Maintenance Ticket Details, Create Downtime Reason, Create Maintenance
Notification Center	Notification List, Notification Details, Notification Preferences
Settings & Profile	Profile, Settings
Optional Enterprise *	Timeline, Audit Logs, Export Center, System Health, Device Management

Mobile Application — 13 Screens (Android & iOS)

MODULE	SCREENS
Authentication	Login, Forgot Password, Reset Password, Change Password
Notification Center	Notification List, Notification Details
Machine Details	Machine Details, Production Summary, Machine Parameters
Alarm & Maintenance	Alarm List, Alarm Details
Profile & Settings	Profile, Settings

* Optional Enterprise screens are not included in the fixed price of ₹15,00,000. They can be taken up as an add-on module and will be quoted separately on request.